

Agent-Based Modeling of *C. elegans*

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Abstract

This project developed and refined an agent-based model of *C. elegans* movement, feeding, and patch interaction over the course of the semester. The work went through three main stages. The first stage used a Laplacian-based growth and diffusion update, which was simple but difficult to keep stable. The second stage moved to a Fourier-space diffusion model with Strang splitting, which made the bacterial update much more stable and accurate. The third stage simplified the system further to focus on worms moving between two patches with timed deposition, making it easier to study behavior without the added complexity of growth and unstable patch dynamics. Along the way, the model was tested with simulation runs, sample movies, and feeding diagnostics. This report summarizes the design changes, the reasoning behind them, and the main outcomes of each iteration.

1 Introduction

C. elegans is a useful model for studying foraging, chemotaxis, and decision-making in patchy environments. In this project, the goal was to build a simple but useful computational model that could reproduce basic worm behavior and help explore how worms respond to food sources over time. The project started with a growth-and-diffusion model for bacteria, then moved through a more stable spectral approach, and finally to a simplified two-patch setup that made it easier to focus on worm behavior itself.

The model development was influenced by previous work on *C. elegans* foraging and collective behavior, especially studies showing that worms switch between exploration and exploitation depending on patch quality and internal state (Haley et al., 2025; Ding et al., 2019). Those papers helped frame this project as a behavioral modeling problem rather than only a numerical simulation problem.

2 Methods

2.1 Method 1: Laplacian Growth and Diffusion

The first version of the model used a finite-difference Laplacian update for bacterial growth and diffusion:

$$b_{\text{new}} = b + dt \cdot (D\nabla^2 b + b(1 - b)). \quad (1)$$

This method was straightforward and easy to implement, but it came with a stability restriction. In practice, the model needed the condition

$$D \cdot \frac{dt}{dx^2} \leq 0.25 \quad (2)$$

to remain stable. When the parameters were too aggressive, the bacterial field could blow up or become numerically unstable. This made the approach useful as a first test, but not ideal for long runs or broader parameter exploration.

2.2 Method 1.2: Gaussian Convolution

The next version replaced the finite-difference diffusion step with Gaussian convolution:

$$b_{\text{new}} = G_{\sigma}(b), \quad \sigma = \sqrt{2Ddt} \quad (3)$$

followed by a growth update:

$$b_{\text{new}} = b_{\text{new}} + dt \cdot b_{\text{new}}(1 - b_{\text{new}}). \quad (4)$$

This removed the main diffusion stability problem because the diffusion step was handled more exactly. It was still an operator-splitting approach, though, so diffusion and growth were not being solved simultaneously. The growth update was still a forward-Euler approximation, so this version was more stable than the Laplacian method but still not fully accurate.

2.3 Method 2: Fourier Space with Strang Splitting

The second major branch used a Fourier-space diffusion update with Strang splitting. The idea was to split the update into half diffusion, full growth, and then half diffusion again:

$$b_{1/2} = \mathcal{F}^{-1} \left(\mathcal{F}(b) \exp(-Dk^2 dt/2) \right), \quad (5)$$

$$b_{\text{grown}} = \frac{Kb}{b + (K - b)e^{-rdt}}, \quad (6)$$

$$b_{\text{new}} = \mathcal{F}^{-1} \left(\mathcal{F}(b_{\text{grown}}) \exp(-Dk^2 dt/2) \right). \quad (7)$$

This version was the most stable and accurate of the early models. It removed the finite-difference restriction from the diffusion step and also improved the growth update by using the analytical logistic solution. In practice, this was the cleanest version for the more general bacterial dynamics.

2.4 Method 3: Simplified Two-Patch Model

After testing the earlier versions, the final branch simplified the system so the focus stayed on worm behavior instead of patch growth complexity. In this version, the model used two patches, point-like worm placement, and timed deposition only. Growth of the

initial patch and deposited bacteria was removed. This made the system easier to run and easier to interpret when studying how worms move, feed, and deposit material over time.

This version was intended as a clearer behavioral model rather than a full bacterial growth model. It allowed the simulations to show the basic worm response to two food sources without the numerical and conceptual complications that came with the earlier growth-and-consumption formulations.

2.5 Simulation and Analysis

The simulation workflow was organized around the main run script, configuration generation, and HDF5 logging. Output was saved as worm history and environment history files so that trajectories, feeding state, and bacterial fields could be analyzed later. The main analysis tools included trajectory plots, movies, feeding diagnostics, and internal consistency checks.

Representative samples from the three model families were stored in the `samples/` folder and used as visual evidence of the development process.

3 Results

3.1 Model Development Over Time

The main result of the semester was not only a final simulation, but a sequence of increasingly usable models. The first Laplacian version was useful for establishing the basic framework, but it was too sensitive to parameter choice. The Gaussian convolution version improved stability, but still left some limitations because growth and diffusion were handled separately. The Fourier/Strang version removed most of the numerical instability and gave a more accurate bacterial update. The final two-patch version then stripped the system down to a simpler setting that was easier to explore and explain.

3.2 Behavior Across Iterations

Across the iterations, the model became easier to run and easier to interpret. The early versions were most useful for testing whether the bacterial dynamics could be represented numerically. The later versions were better for studying worm behavior itself. In the final simplified model, the worms could be compared directly across two patch locations, which made it easier to see how movement, feeding, and timed deposition influenced their paths.

3.3 Representative Outputs

The sample movies show the shift from more technical model testing to more interpretable behavioral simulations. Early samples highlight numerical behavior and patch dynamics, while later samples show cleaner worm trajectories and more controlled interactions with food patches. This progression reflects the main arc of the project: first get the bacterial model working, then improve stability, and finally simplify the setup so worm behavior could be studied more clearly.

Figure 1 shows the initial and final states for the Laplacian model. Figure 2 shows the same comparison for the Gaussian convolution version. Figure 3 shows the Fourier/Strang splitting model, and Figure 4 shows the simplified two-patch model.

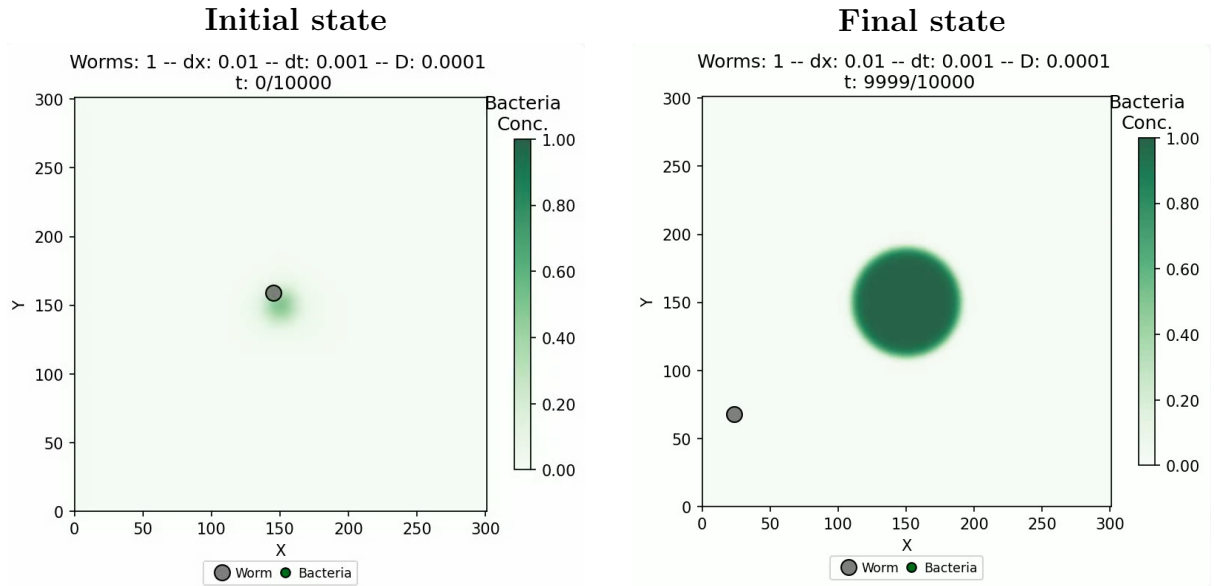


Figure 1: Laplacian model results.

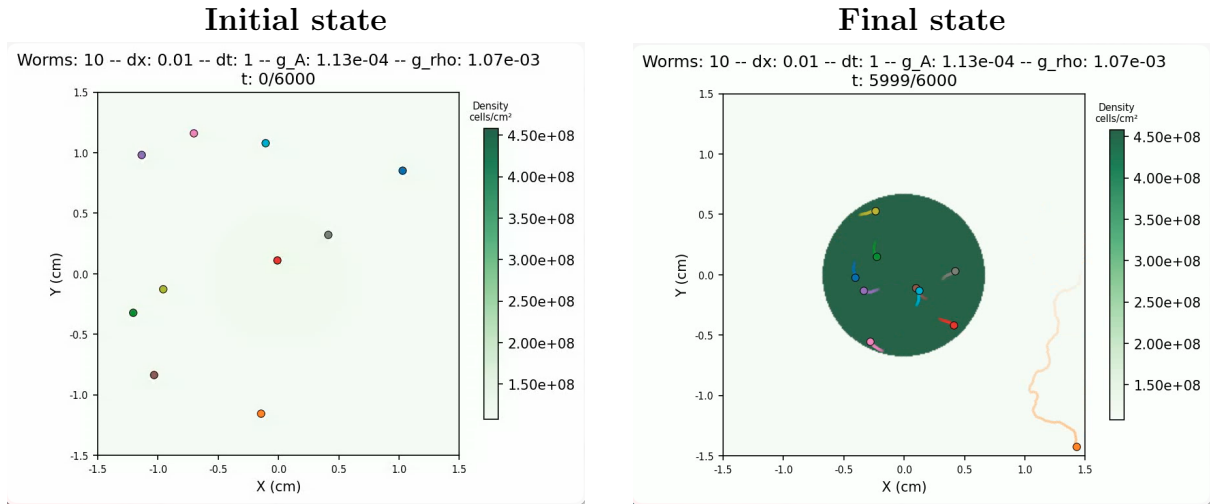


Figure 2: Gaussian convolution model results.

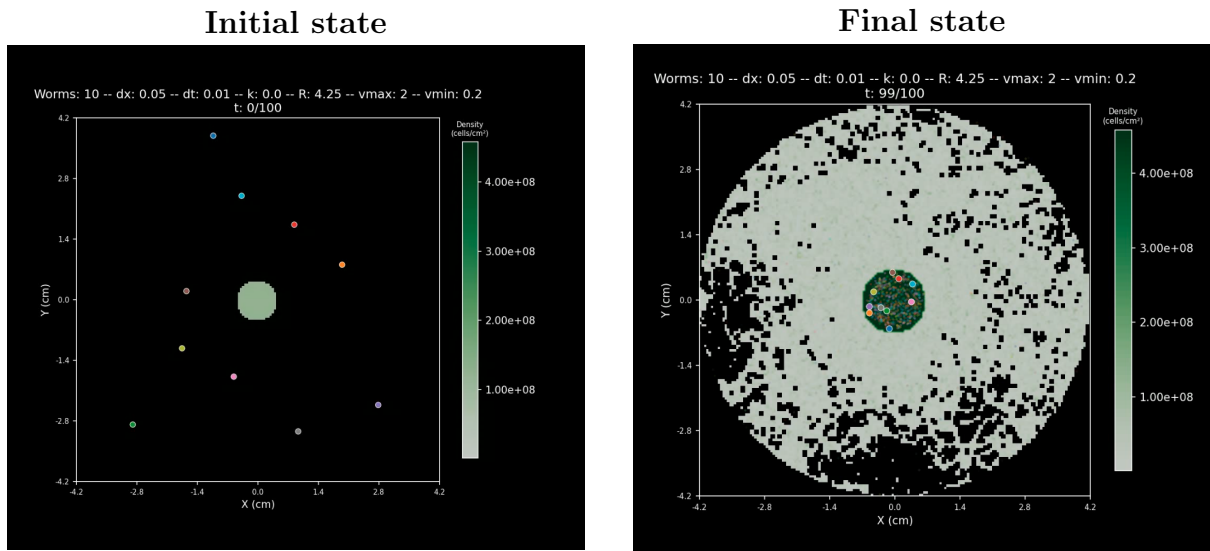


Figure 3: Fourier-space diffusion with Strang splitting.

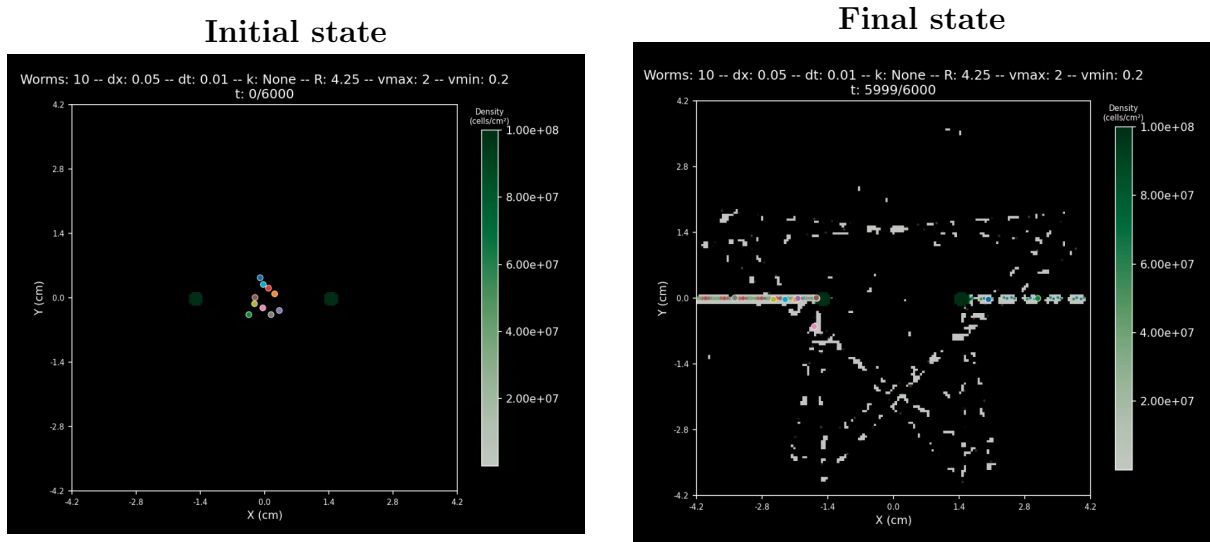


Figure 4: Simplified two-patch model results.

4 Validation

Several checks were used to make sure the model behaved consistently. Feeding logic was tested to confirm that worms only consumed bacteria on patch, that cumulative feeding increased monotonically, and that recorded consumption matched the environment-level bookkeeping. Diagnostic plots were also used to compare feeding and deposition over time.

These checks were important because the project changed direction more than once. Instead of assuming the latest branch was automatically best, the code was verified at each stage so the development path stayed grounded in behavior and not just in numerical convenience.

5 Discussion

The main lesson from this project was that model design and numerical stability are tightly linked. The first method was simple but too restrictive. The second method improved stability and accuracy significantly. The third method made the whole setup easier to study by reducing unnecessary complexity. Taken together, the three methods show a clear progression from a rough first prototype to a more focused behavioral model.

From a biological perspective, the project supports the idea that *C. elegans* foraging can be studied with relatively simple rules, but the exact implementation matters. Small changes in how the bacterial field is updated can have a large effect on what kinds of simulations are practical and what kinds of questions can be asked.

Future work could add back more realism in a controlled way, such as gradually reintroducing growth, more detailed deposition, or worm-to-worm effects once the simpler two-patch model has been fully explored.

Data and Code Availability

The code and data for this project are available at <https://github.com/Mihir-Shrestha/WormABM>.

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